

Lecture 16: Convolutional Neural Networks

Convolution · Pooling · Modern Architectures

Outline

From MLPs to CNNs

2D Convolution

Translation Equivariance, Sparsity, Weight Sharing

Layers, Padding, Stride, Channels

Max / Average Pooling

LeNet, AlexNet, VGG

ResNet, Inception, EfficientNet

Dilated, Transposed, Separable Convolutions

Vision Tasks: Classification, Detection, Segmentation